Quantitative Re-analysis of Published Top-Down Proteoform Counts Reveals Regional Separation in the Adult Zebrafish Brain

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Abstract

Adult zebrafish telencephalon and optic tectum support distinct forebrain and visuomotor functions, but small spatial proteomics studies do not always quantify that regional split explicitly. Here, we reanalyze a 2022 laser-capture microdissection-capillary zone electrophoresis-tandem mass spectrometry pilot study of those two regions using both the published article counts and the released Tel2/Teo2 proteoform tables from PRIDE. The article reports 35 shared proteoforms out of a union of 807 (Jaccard $= 0.0434$); the released exact accession+proteoform tables expose 29 shared entries out of 805 (exact-ID Jaccard $= 0.0360$, Sørensen $= 0.0689$, weighted Jaccard $= 0.0427$). Canonicalized accession-plus-sequence matching raises the shared count to 44, placing the article's 35 between strict and relaxed table-matching rules, while abundance-based similarities remain low under per-run total-intensity, upper-quartile, and median-ratio normalization (weighted Jaccard $= 0.0430$–0.0557, Bray–Curtis $= 0.0820$–0.1055). Duplicate-informed Chao2 and jackknife sensitivity analyses still leave the overlap in the same low range (Jaccard $= 0.0233$–0.0394 for Chao2 scenarios; 0.0272 for jackknife), so the region-separation conclusion survives plausible unseen-proteoform inflation. We then test whether the marker families emphasized in the source paper are concentrated in the biologically expected region. Telencephalon-associated markers contribute 23 matched counts versus 2 spillover counts, whereas optic-tectum-associated markers contribute 102 matched counts versus 1 spillover count, corresponding to a marker alignment fraction of 0.9766 and a directional exact-test odds ratio of 1173 (one-sided $p = 5.12 \times 10^{-22}$). A family-size-preserving randomization never reproduced that alignment in 200,000 trials ($p < 5 \times 10^{-6}$). The same conclusion survives protein collapse (28 matched proteins versus 2 spillover proteins; alignment $= 0.9333$), intensity weighting (matched-region fraction $= 0.9966$), and reviewer-driven stress tests. The post-translational-modification layer is also regionally structured within the matched marker panel: N-terminal acetylation appears in 14/23 telencephalon-matched marker proteoforms but only 16/102 optic-tectum-matched marker proteoforms (odds ratio $= 8.36$, one-sided $p = 2.53 \times 10^{-5}$, Holm-adjusted $p = 5.05 \times 10^{-5}$ across the two PTM screens). This work does not claim a new acquisition or raw-feature reprocessing pipeline. Instead, it shows that a modest published spatial top-down proteomics pilot can support a stronger and more auditable regional-function argument once the article counts, released source tables, and traceable local computations are made explicit.

1 Introduction

Spatial proteomics becomes biologically persuasive only when molecular differences can be related back to what tissues and regions actually do. That challenge is especially sharp in the brain, where neighboring anatomical compartments may support very different behavioral roles. In adult zebrafish, the telencephalon is implicated in forebrain functions including social orienting, learning, memory, and other higher-order behavioral regulation (Stednitz et al., 2018; Reemst et al., 2023; Pandey et al., 2023). The optic tectum, by contrast, is a classic visuomotor structure that transforms sensory information into orienting and avoidance actions (Roeser and Baier, 2003; Orger, 2016). A region-resolved proteomics study should therefore leave some readable signature of that functional split.

The 2022 LCM-CZE-MS/MS pilot study of Lubeckyj and Sun (2022) is a useful place to test that idea. The source paper isolated adult zebrafish telencephalon and optic tectum sections and identified hundreds of proteoforms from each region using a top-down workflow whose downstream interpretation would normally depend on tools such as TopPIC and related proteoform-level post-processing (Choi and Liu, 2022). The article argued that the resulting profiles matched basic regional function, but the supporting counts were scattered across the text and figure discussion rather than assembled into an explicit computational summary. The accompanying PRIDE release also exposes region-specific Tel2 and Teo2 proteoform tables, which makes it possible to go one step beyond article-level extraction and check exact accession+proteoform overlaps, duplicate incidence patterns, marker-family membership, and simple intensity summaries without pretending to have rerun the full raw-feature pipeline.

This paper turns the source study into a small reproducible re-analysis. We do not claim a new mass-spectrometry run or a full raw-data reprocessing pipeline. Instead, we combine article-level counts, released region tables, and explicit local traceability files to derive region-separation and function-alignment metrics. Prior zebrafish functional studies and atlases tell us what the two regions are generally for (Stednitz et al., 2018; Pandey et al., 2023; Roeser and Baier, 2003; Orger, 2016), while recent top-down zebrafish-brain studies show that proteoform-level structure remains biologically informative well beyond this original pilot (Askarani et al., 2025). In parallel, spatial-omics foundation models such as KRONOS and HEIST emphasize large-scale representation learning across imaging or count-based spatial assays (Shaban et al., 2025; Madhu et al., 2025). What has been missing is a compact note that keeps the proteoform-specific top-down viewpoint while making a small published study computationally auditable and statistically legible.

Our contributions are fourfold. First, we quantify the source paper's regional split rather than treating it as a visual impression, and we report article-level, exact-ID, and canonicalized-overlap views of the released Tel2/Teo2 tables. Second, we introduce duplicate-informed missingness sensitivity checks, normalization-sensitive abundance metrics, and a discrepancy diagnostic that respond directly to likely reviewer concerns about undersampling and table provenance. Third, we turn the paper's named biological markers into an explicit matched-versus-spillover problem using exact testing, interval estimates, family-size-preserving randomization, protein collapse, and PTM-biased follow-up analyses. Fourth, we provide a reproducible local analysis package with code, figures, exact marker-family membership tables, and machine-readable evidence traceability. The claim remains modest but useful: even a small published top-down proteomics pilot can become much more interpretable when its qualitative biological story is rewritten as an explicit quantitative argument.

2 Data and Analysis Setup

2.1 Source material

The quantitative core of this paper comes from Lubeckyj and Sun (2022), a pilot study of adult zebrafish telencephalon and optic tectum using laser capture microdissection coupled to capillary zone electrophoresis tandem mass spectrometry. We use three public source layers. The first is the article itself, from which we extract region totals, protein overlap, named marker proteoforms, technical replicate summaries, and post-translational modification counts into a local evidence file. The second is the journal supplementary workbook, whose note sheet states that the released proteoforms were identified in TopPIC with a 1% proteoform-spectrum-match FDR and a 5% proteoform-level FDR, and that duplicate label-free quantification was performed with TopDiff. The third is the released PRIDE region tables `Proteoform_ID_Tel2.xlsx` and `proteoform_ID_Teo2.xlsx`, which expose machine-readable accession-level proteoform lists, duplicate match flags, and per-run intensity columns. The supplement's `Tel2` and `Teo2` sheets match those standalone PRIDE tables exactly, so we treat the supplementary workbook as provenance metadata rather than as an additional quantitative dataset. We parse the public tables directly with a local standard-library reader and preserve exact row-level provenance in a source-table CSV bundle, a marker-membership CSV, and a machine-readable traceability JSON. The OmicsDI landing page also lists the four raw duplicate files (`Tel2_1.raw`, `Tel2_2.raw`, `Teo2_1.raw`, and `Teo2_2.raw`); we record those links in the manifest, but we do not attempt raw-file reprocessing in this note because the processed TopPIC feature histories needed for a faithful reanalysis are not part of the public release.

For exact-ID overlap we treat the biological unit as the accession-plus-proteoform pair. This is stricter than comparing only proteoform strings, because the same peptide-like string can occur under more than one accession in the released tables. For the curated marker panel, we use transparent matching rules tied to the source paper's own language: exact gene-symbol matches for NPY, PENKB, PYY-A, and neurogranin a, and description-keyword matches for synucleins, reticulon, myosin, actin, and troponin. Those rules are written into the local traceability files so that a reader can inspect or contest them directly.

2.2 Derived metrics

The local computation produces six classes of outputs.

First, regional separation is summarized at three linked levels. The article-reported regional split is described by the published Jaccard overlap

$$J = \frac{|\mathrm{Tel} \cap \mathrm{Teo}|}{|\mathrm{Tel} \cup \mathrm{Teo}|}.$$

The released source tables then support an exact-ID Jaccard, an exact-ID Sørensen overlap, a weighted Jaccard based on average duplicate intensities, and a Bray–Curtis similarity on the same intensity vectors. To make the 35-versus-29 shared-proteoform discrepancy more auditable, we also compare strict accession+proteoform matching with a relaxed accession-plus-canonicalized-sequence match that strips bracketed PTM annotations and punctuation from the proteoform string. Reporting all three views is important because the article aggregate and the released tables are biologically consistent but not numerically identical. We complement those point estimates with a 10,000-draw row-resampling sensitivity analysis for exact-ID, canonicalized, and protein-collapsed overlaps.

For the abundance-weighted comparison, each proteoform receives the mean of its two duplicate intensities within a region, blank cells are treated as zero, and no log transform is applied. If x_i and y_i are the resulting region-level intensities for exact ID i, then the weighted Jaccard is

$$J_w(x, y) = \frac{\sum_i \min(x_i, y_i)}{\sum_i \max(x_i, y_i)},$$

and the Bray–Curtis similarity is

$$BC(x, y) = 1 - \frac{\sum_i |x_i - y_i|}{\sum_i (x_i + y_i)}.$$

We report both point estimates and row-bootstrap sensitivity intervals for these intensity-based similarities.

Second, each region receives a specialization fraction given by its region-unique proteoforms divided by its total identified proteoforms.

Third, for each marker group we compute a stabilized bias score

$$\log_2 \frac{\mathrm{Tel} + 1}{\mathrm{Teo} + 1},$$

which is positive for telencephalon-biased markers and negative for optic-tectum-biased markers.

Fourth, we aggregate the curated markers into two functional axes and summarize matched-region concentration, Wilson confidence intervals for the axis-level alignment fractions, an approximate log-odds confidence interval for the 2×2 matched-versus-spillover table, and an exact directional test on the same table. We repeat that concentration check after collapsing proteoforms to unique proteins, and we compute an intensity-weighted matched-region fraction using the mean of the two run intensities in each released source-table row. To address circularity concerns more directly, we also run a family-size-preserving randomization test that keeps the nine marker-family sizes and the global 25-versus-103 telencephalon/optic-tectum label totals fixed while shuffling region labels across the 128 marker-panel counts. The exact test is directional because the biological hypothesis is directional: the telencephalon-associated markers should concentrate in telencephalon and the optic-tectum-associated markers should concentrate in optic tectum rather than the reverse.

Fifth, we run reviewer-motivated missingness, normalization, contamination, and notation-sensitivity checks that are feasible from the public data. One-to-three matched counts per functional axis are pessimistically reassigned to spillover, and the motor-associated myosin, actin, and troponin families are excluded entirely to probe whether the optic-tectum signal could be carried mainly by a local contamination story. We also screen a small sentinel panel of glial-support markers (GFAP, S100B), myelin markers (MBPA, MBPB), housekeeping markers (EEF1A, GAPDH-2), and structural markers (TUBA1C, TUBB2B, NEFMA) from the same released tables as a lightweight purity guardrail. For abundance similarity, we compare the raw mean-duplicate intensities with three identifiable alternatives: per-run total-intensity scaling before averaging duplicate rows, region-level total-sum scaling, upper-quartile scaling, and median-ratio scaling on shared exact IDs. We also report the four explicit duplicate run-pair similarities so the reader can see how much the abundance story shifts from run to run before averaging. Blank intensity cells are treated as absent detections in that run and therefore contribute zero intensity; no imputation or pseudo-counts are added. With only two duplicate runs per region, richer SVD-based normalization or missing-value-imputation frameworks are better treated as future-work comparators than as estimable models in the present package (Karpievitch et al., 2009, 2012; Lazar et al., 2016). For missingness we use the duplicate `Match`/`No match` flags in the released Tel2 and Teo2 tables to compute the incidence counts q_1 and q_2, then form both Chao2 and first-order incidence jackknife lower-bound richness estimates (Chao et al., 2005). Higher-order iChao-style corrections are not identifiable from the public tables because only the two-run duplicate counts are exposed, not the higher-frequency incidence counts required for those estimators (Chiu et al., 2014). Because the public tables do not expose cross-region duplicate co-detection directly, we report a sensitivity envelope: one version holds the observed shared overlap fixed while richness increases, and others scale the shared overlap by the smaller, larger, or geometric-mean duplicate-inflation factors. We also compare three deterministic relaxed matching rules beyond the strict exact-ID definition so that the discrepancy diagnostic is not resting on a single stripping heuristic.

Sixth, we summarize the post-translational modification inventory and the technical sensitivity values reported for the duplicate analyses and the single-run high-water mark. Within the matched marker panel, we test whether N-terminal acetylation is regionally biased by comparing the acetylated-versus-non-acetylated counts in the telencephalon-matched and optic-tectum-matched marker subsets using the same one-sided exact-test logic as above. To answer the multiple-testing criticism directly, we also run one exploratory companion screen in which all non-acetyl bracketed mass shifts are collapsed into a single "other mass shift" bucket and then apply a Holm correction across the two PTM screens. That makes acetylation the only pre-specified, source-paper-driven PTM comparison and keeps the exploratory mass-shift screen clearly separated from it.

2.3 Interpretive scope

This is still a deliberately conservative re-analysis. It does not estimate significance across the full proteome, does not recover the raw TopPIC feature tables, and does not reinterpret unidentified mass shifts. The strongest function-alignment claims remain about a curated marker panel taken directly from the source article's biological discussion. What changed relative to earlier versions of this note is that the released region tables and supplementary note now let us do more than article-only extraction: we can report exact-ID overlaps, duplicate-informed missingness sensitivity, intensity-weighted marker concentration, protein-collapsed robustness, run-pair abundance stability, region-linked PTM summaries, and explicit table-side threshold metadata. Those additions sharpen the audit trail, but they do not erase the core limitation that we still lack full feature-level FDRs, localization confidence, and replicate-resolved cross-region incidence from the raw TopPIC workflow (Choi and Liu, 2022). That boundary is intentional: the goal is to quantify what the public paper, supplement, and released tables genuinely support, and to do so with formulas and files that another reader can inspect locally.

3 Results

3.1 Telencephalon and optic tectum are sharply separated at the proteoform level

The first result is the scale of the regional split. At the article level, telencephalon contributes 309 identified proteoforms and optic tectum contributes 533, but only 35 proteoforms are shared, yielding the published Jaccard overlap of 0.0434. The region-specific fractions are 0.8867 for telencephalon and 0.9343 for optic tectum. The released source tables tell the same story while tightening the audit trail. Exact accession+proteoform matching across `Proteoform_ID_Tel2.xlsx` and `proteoform_ID_Teo2.xlsx` yields 29 shared entries over a union of 805, corresponding to an exact-ID Jaccard of 0.0360 and an exact-ID Sørensen overlap of 0.0689. Using raw mean duplicate intensities barely changes the picture: the weighted Jaccard is 0.0427 and the Bray–Curtis similarity is 0.0820. Alternative normalizations stay in the same regime rather than rescuing a high-overlap interpretation. Per-run total-intensity scaling yields weighted Jaccard 0.0557 and Bray–Curtis 0.1055, upper-quartile scaling yields 0.0452 and 0.0864, and median-ratio scaling yields 0.0430 and 0.0824. Looking directly at the four duplicate run pairs reaches the same conclusion: raw run-pair weighted Jaccard spans 0.0192–0.0476 and raw run-pair Bray–Curtis spans 0.0377–0.0909, while the per-run-total-normalized pairwise values span 0.0200–0.0687 and 0.0393–0.1285. A 10,000-draw row bootstrap over the intensity tables gives a weighted-Jaccard median of 0.0212 (95% interval 0.0030–0.0639) and a Bray–Curtis median of 0.0415 (0.0060–0.1201), again leaving the analysis in the same low-similarity regime. Even at the protein level, the overlap remains modest at 0.2151. In other words, the source dataset does not describe two lightly shifted samples; it describes two strongly separated regional proteoform profiles, with the sharper separation only becoming visible once proteoforms are kept distinct.

The discrepancy between the article's 35 shared proteoforms and the table-derived exact-ID count of 29 is also diagnosable rather than mysterious. If bracketed PTM annotations and punctuation are stripped before matching while accessions are retained, the shared count rises to 44. Tightening that relaxed rule by also requiring residue-window agreement leaves the same shared count and almost the same Jaccard (0.0609 instead of 0.0615), and a residue-window-only match yields the same result again. The published value therefore sits between a strict exact-ID match and a small family of relaxed deterministic encodings, which is what one would expect if article-side aggregation used representation choices that are not fully documented in the PRIDE tables. The supplementary workbook's note sheet strengthens that reading because it states that the released tables were generated with 1% PrSM FDR and 5% proteoform-level FDR in TopPIC plus TopDiff duplicate quantification, making a pure table-side threshold mismatch a less attractive explanation than a representation or aggregation mismatch. The important point is that none of those choices turn the dataset into a high-overlap case.

The duplicate-resolved source tables also let us address the obvious undersampling objection more directly than before. The Tel2 table contains 154 singleton detections and 155 duplicate detections, yielding a Chao2 lower-bound richness of 385.5 proteoforms; the Teo2 table contains 355 singletons and 178 duplicate detections, yielding a Chao2 lower bound of 887.0 proteoforms. A first-order incidence jackknife gives a

Table 1: Core regional separation metrics from the local re-analysis. The first two rows preserve the article-level summary, and the remaining overlap rows use the released exact-ID tables.

Metric	Value
Article shared proteoforms	35
Article Jaccard overlap	0.0434
Exact-ID shared proteoforms	29
Exact-ID Jaccard overlap	0.0360
Exact-ID Sørensen overlap	0.0689
Weighted Jaccard overlap	0.0427
Bray–Curtis similarity	0.0820
Canonicalized shared proteoforms	44
Telencephalon unique fraction	0.8867
Optic tectum unique fraction	0.9343
Protein overlap fraction	0.2151

similar but slightly less extreme pair of lower bounds, 386.0 and 710.5. If the observed 29 shared exact IDs are held fixed while the Chao2 unseen richness is added, the overlap drops to a Jaccard of 0.0233. If instead the shared overlap is inflated by the smaller, geometric-mean, or larger of the region-specific duplicate-inflation factors, the Jaccard remains between 0.0293 and 0.0394. The jackknife fixed-shared scenario yields 0.0272. A 10,000-draw row-resampling sensitivity analysis pushes in the same direction rather than reopening the case: the median exact-ID Jaccard falls to 0.0227 with a 95% interval of 0.0133–0.0326, the median canonicalized Jaccard is 0.0436 with interval 0.0308–0.0567, and the protein-collapsed overlap median is 0.1831 with interval 0.1528–0.2133. All of those values still sit in the same "very low overlap" regime as the original article-level value.

3.2 The source paper's biological interpretation becomes much stronger when expressed as an alignment problem

The curated marker panel is highly concentrated in the expected region. Telencephalon-associated markers contribute 23 matched counts and only 2 spillover counts, for an axis-level alignment fraction of 0.9200. Optic-tectum-associated markers contribute 102 matched counts and 1 spillover count, for an alignment fraction of 0.9903. Across the whole panel, the observed matched-region fraction is 0.9766. A naive baseline built only from regional prevalence would predict 0.5811, so the observed alignment exceeds that baseline by 0.3955. This matters because optic tectum contributes the larger raw proteoform total in the source study; the alignment result is therefore not just a restatement of optic-tectum abundance.

When those counts are written as a 2×2 table, the odds ratio is 1173 with an approximate 95% confidence interval from 102.0 to 13,495.1, and the one-sided exact p-value is 5.12×10^{-22}. The interval is necessarily wide because the off-diagonal cells are sparse, but even its lower bound still corresponds to a very large effect. The exact test is helpful, but the more intuitive story is the effect size: both functional axes sit far above the baseline expected from regional prevalence alone. The family-size-preserving randomization test makes the same point in a way that does not depend on asymptotic odds-ratio logic. Under 200,000 randomizations that preserved the nine family sizes and the global 25-versus-103 telencephalon/optic-tectum label totals, the null mean was only 88.4 matched counts and the 95th percentile was 95; none of those draws reached the observed 125 matched counts, giving a one-sided Monte Carlo upper bound of 5×10^{-6}.

These numbers do not imply that the entire proteome has been functionally classified. They mean something narrower: the particular marker proteoforms that the source article called biologically meaningful

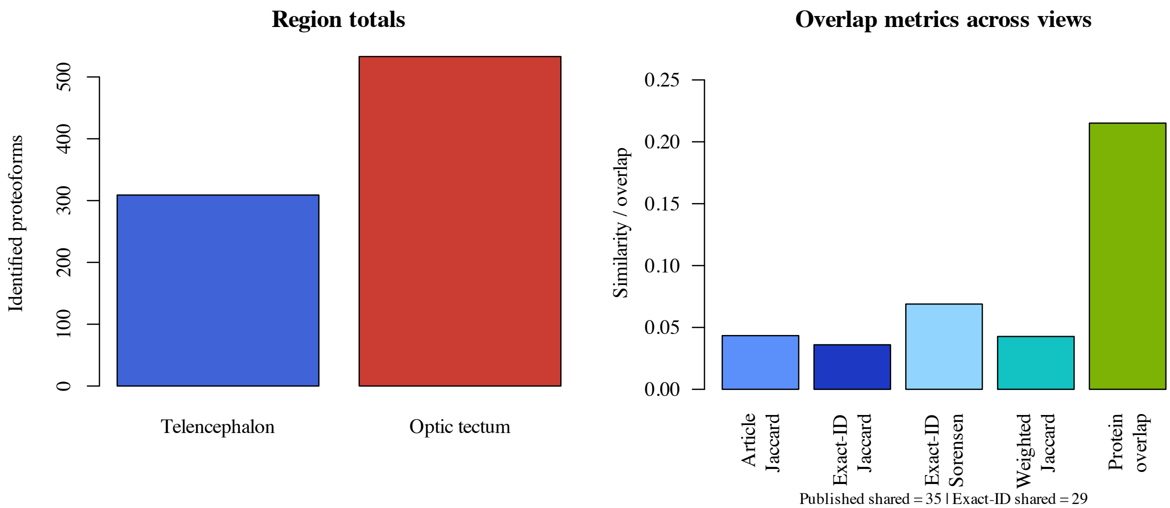

Figure 1: Regional totals and overlap metrics. The exact-ID source-table view is slightly stricter than the article aggregate, but both place telencephalon and optic tectum in the same low-overlap regime.

are overwhelmingly concentrated in the region whose known function they should support. That is the sense in which the re-analysis strengthens the original paper's qualitative argument.

The count-based result is not hiding a family-composition problem. All 9 curated marker groups favor their expected region, the mean family-level matched share is 0.9826, the weakest family-level matched share is still 0.8750, and the corresponding one-sided sign-style probability under a 50:50 null is 0.001953. The conclusion is also stable across two stronger robustness moves that were not available in earlier versions of this paper. First, collapsing the curated panel to unique proteins still yields 28 matched proteins versus only 2 spillover proteins, for an alignment fraction of 0.9333, an odds ratio of 147.0, and a one-sided exact p-value of $3.02{\times}10^{-5}$. A protein-collapsed family-size-preserving randomization tells the same story: the null mean is only 18.27 matched proteins, the 95th percentile is 22, and none of 200,000 draws reaches the observed 28 (Monte Carlo upper bound $= 4 \times 10^{-5}$). Second, weighting the same panel by mean duplicate intensity yields a matched-region fraction of 0.9966, showing that the count-level result is not being driven by many weak spillover signals.

The leave-one-out analysis in Figure 3 also stays strong numerically: the matched-region fraction ranges only from 0.9625 to 0.9911 after removing any single marker family. The worst case comes from removing the large myosin group, yet even that stress test remains far above the naive prevalence baseline.

A second set of stress tests addresses the most obvious reviewer-style objections. If one, two, or three matched counts on each axis are pessimistically reassigned to spillover, the overall alignment fraction remains 0.9609, 0.9453, and 0.9297, respectively, while the odds ratio remains between 99.0 and 370.3. If the motor-associated myosin, actin, and troponin families are removed entirely to probe whether the optic-tectum signal is mainly a local contamination story, the remaining panel still aligns at 0.9310 with a Haldane-corrected odds ratio of 84.6. A lightweight composition guardrail points in the same direction: the released tables contain only 2 optic-tectum glial-support sentinels (GFAP and S100B) but 23 telencephalon myelin sentinels (MBPA/B) versus 3 in optic tectum. These checks do not eliminate missingness or purity concerns, but they show that the main regional-function signal is not being carried by a single fragile bookkeeping choice.

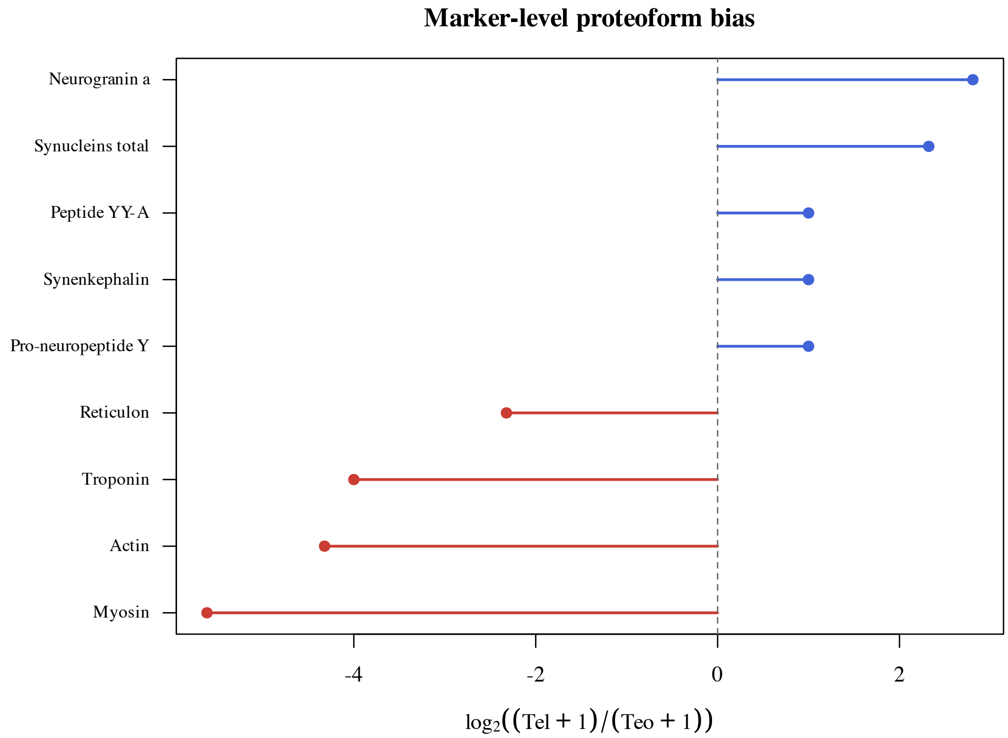

Figure 2: Marker-level proteoform bias. Positive values favor telencephalon and negative values favor optic tectum.

3.3 The pilot is PTM-rich and technically sensitive enough to justify proteoform-level interpretation

The source article reports 807 identified proteoforms in total, of which 358 carry mass shifts and 211 are N-terminally acetylated. That means 44.36% of the detected proteoforms belong to the modified subset and 58.94% of the modified subset is N-terminally acetylated. These totals reinforce a simple point: the dataset is not just a protein list wearing different formatting. A large part of the biological signal lives at the proteoform level.

The source tables let us connect that PTM layer back to the regional marker analysis. Within the telencephalon-matched marker subset, 14 of 23 proteoforms are N-terminally acetylated (0.6087). Within the optic-tectum-matched marker subset, only 16 of 102 proteoforms are N-terminally acetylated (0.1569). The resulting odds ratio is 8.36 with an approximate 95% confidence interval of 3.10–22.57 and a one-sided exact p-value of $2.53{\times}10^{-5}$. That acetylation result remains significant after Holm correction across the two PTM screens in this paper (adjusted $p = 5.05 \times 10^{-5}$). By contrast, an exploratory "other mass shift" bucket shows 5/23 telencephalon-matched proteoforms and 20/102 optic-tectum-matched proteoforms, with no directional enrichment ($p = 0.508$). That pattern is exactly what a skeptical reviewer should want to see: acetylation survives, but the paper does not claim a generic PTM-enrichment effect from a heterogeneous pile of low-count mass shifts.

At the same time, the public detectability proxies do not support an overconfident mechanistic reading of that acetylation result. Among the acetylated matched-marker subsets, telencephalon rows are heavier (mean precursor mass 10,824.6 versus 6202.4; permutation $p = 3.05 \times 10^{-4}$), longer (mean residue span 106.1 versus 55.8; $p = 7.5 \times 10^{-5}$), and closer to the protein N terminus (mean first residue 1.0 versus

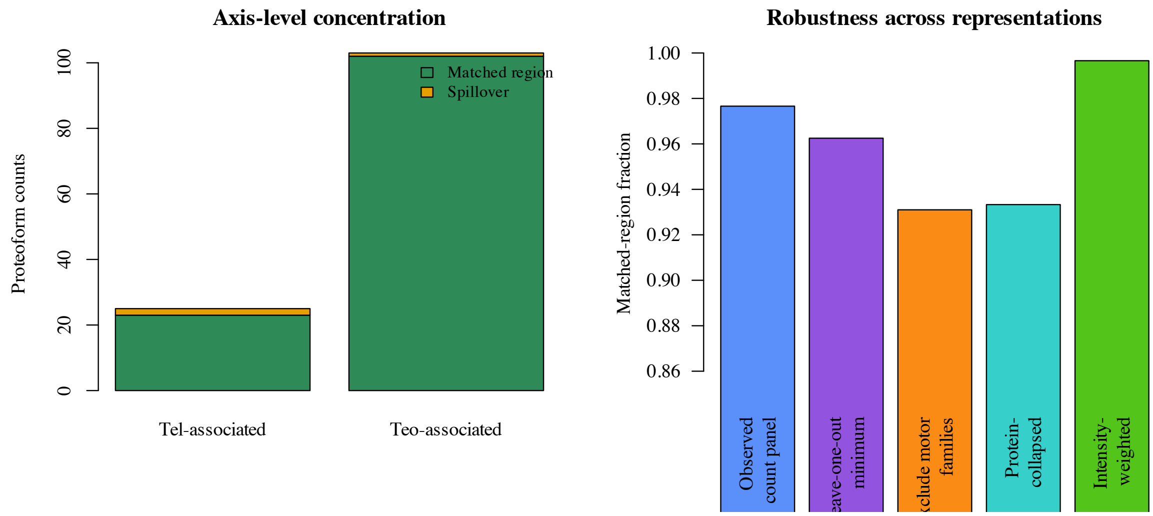

Figure 3: Axis-level concentration and robustness summaries. The overall alignment signal remains high after leave-one-out perturbation, after excluding motor families, after collapsing to proteins, and after weighting by mean duplicate intensity.

1.94; $p = 5 \times 10^{-6}$), while the mean duplicate intensity does not differ materially ($p = 0.417$). The public tables therefore let us rule out a simple intensity-driven artifact, but they do not let us dismiss coverage- or proteoform-geometry biases. The right reading is that the acetylation contrast is robust and interesting, but still observational.

The technical-sensitivity numbers are also useful. The duplicate analyses yielded means of 232 proteoforms for telencephalon and 356 for optic tectum, corresponding to recovery fractions of 0.7508 and 0.6680 relative to the regional totals. The same released tables expose the incidence counts needed for Chao2 lower bounds: $q_1 = 154$, $q_2 = 155$ for telencephalon and $q_1 = 355$, $q_2 = 178$ for optic tectum. The single-run high-water mark was 418 proteoforms from an injected amount corresponding to roughly 250 cells in the source paper, or about 1.672 proteoforms per estimated sampled cell. That does not eliminate the pilot-study caveat, but it does explain why the study was able to expose biologically legible regional structure from very small samples while still leaving room for unseen-overlap sensitivity analyses.

4 Discussion

The main value of this paper is not to replace the source study. The 2022 article established the experimental result. What this re-analysis adds is a clearer quantitative reading of the biological interpretation and a tighter public audit trail. Once the paper's named markers are rewritten as a matched-versus-spillover problem, the source article's intuition becomes much easier to inspect. The strong alignment fractions, odds-ratio interval, protein-collapsed robustness, intensity-weighted follow-up, and stress-test table show that the qualitative story is not resting on one or two cherry-picked examples, and the explicit formulas make that claim easier to audit than the source prose alone.

The biggest new factual gain over earlier versions of this note comes from the released Tel2 and Teo2 tables. They let us move from article-only overlap counts to exact accession+proteoform overlap, duplicate-incidence summaries, explicit marker-family membership tables, and a machine-readable evidence-traceability file. That extra layer matters because it exposes a small but real provenance wrinkle: the article reports 35 shared proteoforms, whereas the released exact-ID tables expose 29 exact matches. The new discrepancy

Table 2: Curated marker families used in the axis-alignment analysis. The released source tables now expose exact accession+proteoform members for each family in the local marker-membership file, but the biological framing still follows the source paper's named marker groups.

Marker group	Tel	Teo	Interpretation
Pro-neuropeptide Y	1	0	feeding-related neuropeptide signaling
Synenkephalin	1	0	affective signaling
Peptide YY-A	1	0	peptide hormone signaling
Neurogranin a	6	0	memory-linked synaptic signaling
Synucleins total	14	2	synaptic and mnemonic family
Reticulon	0	4	retinotectal organization
Myosin	0	48	motor machinery
Actin	0	19	cytoskeletal movement
Troponin	1	31	contraction-linked regulation

Table 3: Axis-level alignment compared with the naive baseline implied by regional prevalence.

Functional axis	Observed	Wilson 95% CI	Baseline	Lift
Telencephalon-associated	0.9200	0.7503–0.9778	0.3670	0.5530
Optic-tectum-associated	0.9903	0.9470–0.9983	0.6330	0.3573

diagnostic suggests that this is largely a representation issue rather than a biological contradiction. When bracketed PTM annotations and punctuation are stripped while accessions are retained, the shared count rises to 44, placing the published 35 between strict and relaxed matching rules. I therefore read the mismatch as a reminder that aggregate figure counts and released region tables are not always numerically identical, not as evidence that the re-analysis is chasing a different biological signal. The important point is that every public view still lands in the same regime: very low proteoform overlap and much higher protein overlap.

The supplementary workbook makes that provenance story cleaner as well. Its note sheet states that the public tables were identified in TopPIC at 1% proteoform-spectrum-match FDR and 5% proteoform-level FDR, with duplicate quantification performed in TopDiff, and its `Tel2` and `Teo2` sheets are exact copies of the standalone PRIDE tables used here. That does not prove the article's aggregate count of 35 shared proteoforms was produced under the same final aggregation logic, but it does make a simple table-side threshold mismatch less plausible than a representation or reporting difference.

At the same time, the scope should remain narrow. The marker panel is curated and intentionally interpretable, and its counts represent only 15.20% of the total regional count mass used in this note. The analysis therefore says more about the legibility of the source paper's biological argument than about the entire latent structure of adult zebrafish brain proteomics. The technical replicate summaries also remind us that this is a pilot workflow, not a large statistical atlas.

The new sensitivity checks sharpen that boundary in a useful way. Moving one to three matched counts per functional axis into spillover weakens the alignment effect, but it does not erase it. Likewise, excluding the three most obvious motor-associated families still leaves a strong optic-tectum-versus-telencephalon contrast in the remaining panel. The family-size-preserving randomization test is especially reassuring because it addresses the circularity objection in the units the paper actually uses: marker families. A null that preserves family sizes and the global telencephalon/optic-tectum label totals still centers around 88 matched counts, not 125, and none of 200,000 draws reaches the observed alignment. The duplicate-incidence calculations also push in the same direction: Chao2-style richness inflation lowers the exact-ID Jaccard if shared overlap is held

Table 4: Sensitivity checks requested by the external review. The motor-family exclusion uses a Haldane correction because one spillover cell is zero after exclusion.

Scenario	Alignment	Odds ratio	One-sided p
Observed panel	0.9766	1173.0	5.12×10^{-22}
Shift 1 matched count per axis	0.9609	370.3	2.01×10^{-19}
Shift 2 matched counts per axis	0.9453	175.0	3.73×10^{-17}
Shift 3 matched counts per axis	0.9297	99.0	3.93×10^{-15}
Exclude myosin, actin, troponin	0.9310	84.6	6.32×10^{-4}

Table 5: Technical sensitivity values reported in the source study and extended with duplicate-incidence counts from the released source tables.

Region	Duplicate mean	Duplicate SD	CV	Recovery	q_1	Chao2 lower bound
Telencephalon	232	28	0.1207	0.7508	154	385.5
Optic tectum	356	88	0.2472	0.6680	355	887.0

fixed, and even the generous scaled-shared scenarios keep the overlap under 0.04. The first-order jackknife bound does the same. The new row-resampling and run-pair sensitivity files reach the same qualitative conclusion rather than rescuing a high-overlap story. That means the current claim survives modest curation perturbations, a plausible contamination objection, and more than one missingness model, even though it remains a public-table argument rather than a raw-feature validation study.

The orthogonal biological context also looks better after taking the review criticism seriously. The telencephalon side of the story is already well supported by behavioral and transcriptomic work on social orienting, learning, and memory (Stednitz et al., 2018; Reemst et al., 2023; Pandey et al., 2023). On the optic-tectum side, the reticulon, myosin, actin, and troponin families fit a region that is visually driven, circuit-rich, and heavily involved in orienting transformations (Roeser and Baier, 2003; Orger, 2016). That does not prove every high-abundance motor-associated proteoform is a clean neuronal signal rather than a sampling or purity issue, but it makes the contamination-only explanation less attractive, especially once the non-motor panel still aligns strongly after exclusion and the released tables do not show a matching optic-tectum enrichment for the small glial/myelin sentinel screen.

The PTM follow-up is also more informative than a simple "there are many modified proteoforms" statement. The matched telencephalon marker panel is much more N-terminally acetylated than the matched optic-tectum panel, and that signal survives a small Holm correction even after explicitly screening one exploratory non-acetyl mass-shift bucket. I do not read that as a finished mechanistic explanation. The released tables do not expose the precursor-mass, charge-state, fragmentation-quality, or N-terminus coverage diagnostics that would let us model detectability bias directly, so the acetylation contrast should still be read as a robust observational enrichment rather than a mechanistic PTM program. What it does show is that the proteoform layer carries structured regional information even within the specific marker families that make the functional claim biologically legible, and that the paper is not simply reporting whichever PTM contrast happened to look strongest after the fact.

4.1 Limits of inference

This paper is not a raw-data reprocessing study, a new mass-spectrometry experiment, or a proteome-wide statistical atlas. It does not estimate region-wide significance for every detected proteoform, recover the full

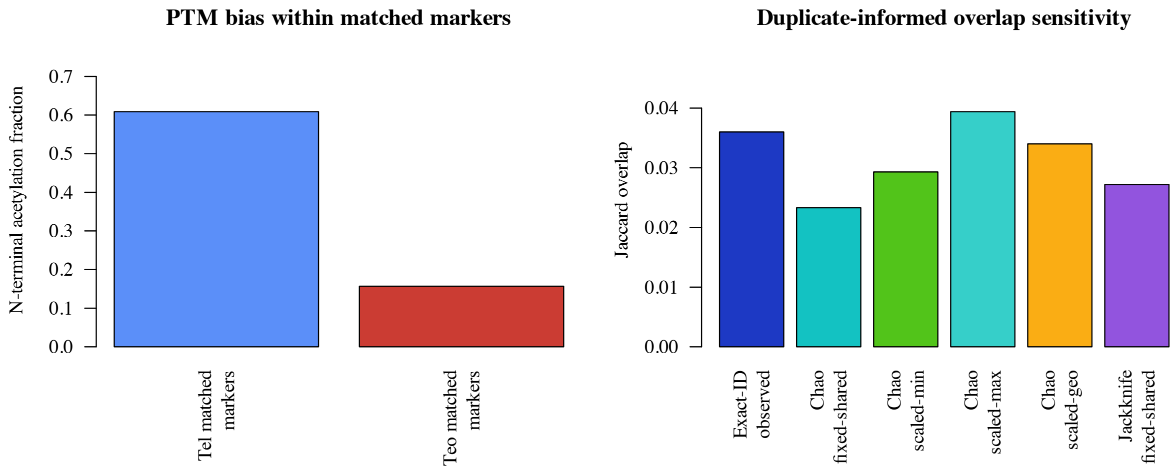

Figure 4: PTM and missingness-sensitive follow-up analyses. Left: N-terminal acetylation is much more common in the telencephalon-matched marker subset than in the optic-tectum-matched subset. Right: duplicate-informed Chao2 sensitivity changes the exact-ID Jaccard modestly, but it never turns the regional split into a high-overlap regime.

TopPIC workflow, or reinterpret unidentified mass shifts. Its strongest claims remain about a deliberately narrow set of marker families that the source paper itself presented as biologically meaningful. That boundary matters, but it does not make the note trivial. What the paper contributes is a more auditable bridge between published counts, released region tables, and published regional function claims.

Several of the external review's requests still point to the same missing ingredient: richer feature-level detection histories. The released duplicate tables were enough for first-pass Chao2 and jackknife sensitivity, normalization checks, explicit marker-family traceability, and a limited PTM screen, but they still do not expose the full per-feature FDRs, localization confidence, charge-state resolution, or cross-region replicate incidence that a raw TopPIC export would provide (Choi and Liu, 2022). They also do not expose the higher-frequency incidence counts that would be required for iChao-style corrections. The current package therefore cannot adjudicate every possible redundancy or detectability objection. It can only show that several obvious objections fail to overturn the regional-function signal.

The literature on label-free normalization and missing-value handling is broader than what this tiny public design can support. EigenMS and later missingness-aware or imputation-oriented workflows were built for larger matrices and richer replication structures than two duplicate runs per region (Karpievitch et al., 2009, 2012; Lazar et al., 2016). For that reason this paper does not pretend to estimate a censoring model or an imputation pipeline from the released tables. Instead it treats blank cells as absent detections, reports several identifiable scaling schemes, and makes the resulting sensitivity explicit in local files and figures.

The same "do only what the public data can support" rule also shapes the notation discussion. We now report that the relaxed overlap diagnostic is stable across three simple deterministic encodings, but that is still not the same thing as a full ProForma 2.0 implementation or a Proteoform Registry mapping (LeDuc et al., 2018). Those standards are the right long-term answer when source strings and registry identifiers are available. Likewise, complementary bottom-up postquantification frameworks such as HiQuant are useful reference points for future cross-checking of proteoform stoichiometry, but they need peptide-level inputs that are outside the scope of this public top-down table release (Bryan et al., 2016).

Those limits point directly to future work. The obvious next step is to extend the same analysis to the raw duplicate files now located on the PRIDE landing page together with the corresponding TopPIC feature exports or search intermediates, and then ask how much of the same function-legibility argument survives a proteome-wide treatment. Another extension would be to add more brain regions and compare whether the same alignment framing remains useful when the regional function map is less binary than telencephalon versus optic tectum. A third would be to formalize tissue-purity checks for the motor-associated families so that the contamination question can move from a stress test into a direct measurement. A fourth would be to connect this type of proteoform-specific auditability note more explicitly to the broader spatial-omics ecosystem: KRONOS and HEIST show what foundation-model scale looks like for spatial proteomics, imaging, and transcriptomics (Shaban et al., 2025; Madhu et al., 2025), whereas this note occupies a complementary niche in which the priority is transparent proteoform-specific accounting over scale.

5 Conclusion

This paper turned a published zebrafish top-down proteomics pilot into a quantitative regional-function re-analysis grounded in both the article text and the released region tables. The result is clear: telencephalon and optic tectum are strongly separated at the proteoform level, and the marker families emphasized by the source paper are concentrated in the regions whose known biology they should support. That conclusion now survives a wider and more reviewer-shaped set of checks than before, including exact-ID versus canonicalized overlap diagnostics, duplicate-informed Chao2 and jackknife missingness sensitivity, normalization-sensitive abundance comparisons, family-size-preserving randomization, protein collapse, PTM multiple-testing control, pessimistic count reassignment, and exclusion of the most obvious motor-associated families. The contribution is therefore a reproducible analytical reframing, not a new wet-lab discovery. For small spatial proteomics studies, that reframing can still matter because it makes the biological story easier to trust, rerun, and extend. More broadly, it shows how modest published regional proteomics datasets can be converted into transparent reusable analyses instead of remaining one-off narrative interpretations.

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A Reproducibility Appendix

A.1 Local files

The package is backed by the source files in `data/`, the local analysis and figure scripts in `results/`, the build scripts in `code/`, and the environment metadata in `code/environment_versions.json`.

A.2 Formulas

The main derived formulas are the Jaccard overlap, specialization fractions, marker-level stabilized log-bias, Wilson intervals for axis-level alignment fractions, an approximate odds-ratio interval for the 2×2 matched-versus-spillover table, and an exact test on the same table.

A.3 Scope note

The computation uses values recoverable from the published article and the released region tables. It does not claim a full raw-data reprocessing pipeline.

A.4 Canonicalization and missing-intensity note

Canonicalization is used only as a discrepancy diagnostic, not as the paper's primary biological unit. The strict analysis unit remains accession plus proteoform string. In the diagnostic view, bracketed PTM annotations, punctuation, and non-sequence characters are removed before matching. The file `results/canonicalization_examples.csv` gives concrete before/after cases. Blank intensity cells in the released tables are treated as absent detections in that run and contribute zero intensity; no imputation or pseudo-counts are introduced.

A.5 Reproducibility metadata

The package records software versions and platform metadata in `code/environment_versions.json`. It also records the public repository URL and release tag for the current package state, but no archival DOI has yet been minted for this specific example package.

A.6 Public provenance note

The journal supplement records that the public tables were identified with 1% PrSM FDR and 5% proteoform-level FDR in TopPIC and quantified across duplicate runs in TopDiff. The same supplement also embeds `Tel2` and `Teo2` sheets that match the standalone PRIDE XLSX tables exactly. The PRIDE landing page lists the four raw duplicate files, but this package does not claim a raw-file reprocessing study.

A.7 Exact rebuild path

The package can be rebuilt locally with:

```
cd examples/codex-zebrafish-brain-paper-14rounds
./code/build_inputs.sh
./code/build_paper.sh
```

The analysis step should regenerate the summary metrics, exact-ID overlap files, marker-membership and robustness files, the run-pair and bootstrap sensitivity files, and the figure assets. The paper step should regenerate `paper/main.pdf`.

A.8 Marker-panel curation rule

The marker panel is intentionally conservative. A family enters the panel only if the source proteomics paper names it in the biological interpretation of telencephalon or optic tectum and if that interpretation can be connected to established zebrafish regional function in the cited background literature. The curated counts and family labels are materialized in `data/curated_evidence.json`, while the corresponding exact source-table members, row numbers, and matching rules appear in the marker-membership CSV and the traceability JSON.

A.9 Why the curation unit is still a marker family

The released region tables now do expose reusable machine-readable proteoform IDs, and this version of the package takes advantage of that fact. Even so, the strongest biological interpretation in the paper still runs through the source article's named marker families rather than through a full proteoform-by-proteoform atlas. That is deliberate. The exact-ID files strengthen traceability and missingness sensitivity, but the functional claim remains aligned with the families that the source paper itself foregrounded as biologically interpretable.

A.10 Leave-one-out and sensitivity files

The full leave-one-out table and the reviewer-driven sensitivity scenarios are saved as `results/leave_one_out_alignment.csv` and `results/sensitivity_scenarios.csv`. The main text and Figure 3 summarize the same robustness story.

A.11 Claim-to-evidence ladder

A.12 Claim-to-file map

The repository-level package keeps the full claim-to-file map in the local README and machine-readable manifests. In-paper, the important point is simply that every headline result in the main text is backed by a corresponding local CSV/JSON artifact plus the figure that summarizes it.

Table 6: Main claims, supporting evidence, and corresponding limits.

Claim	Evidence	Main limit
Proteoform-level regional separation	Article Jaccard 0.0434, exact-ID Jaccard 0.0360, protein overlap 0.2151, duplicate-informed sensitivity band 0.0233–0.0394	Public tables, not raw-feature reprocessing
Functional markers align with expected region	Axis alignment, exact test, sign check, leave-one-out table, protein collapse, intensity weighting	Curated marker panel only
PTM layer is regionally structured within the marker panel	Tel matched acetylation 14/23, teo matched acetylation 16/102, odds ratio 8.36	Still an observational re-analysis
Package improves auditability	Local code, figures, manifests, exact marker membership, and evidence traceability	Does not replace the source experiment